The objects we know as "Listeners" are actually controllers.

MVC is the basis for Swing's Look and Feel. Another advantage of MVC is that any View is automatically updated when anything changes in the Model (data) it is based on. Finally, because the Model is independent of the Views and Controllers, any changes to the Views or Controllers do not affect the underlying Model.

An Example- Button:

The Button is a model—it keeps its name and caption(its Data). The Button doesn't decide how to draw itself—the View decides that using a Layout manager. And neither does The Button to react to being pressed—the Controller decides how to handle that. As you might imagine, the upshot of this design is flexibility. When the data value of a model changes, the Controller hears about it. The Controller [Listener] is notified of the change. An event happens, and the Listener hears about the event.

There are two Kinds of Notification

- Lightweight Notification - Minimal information is passed. This kind of notification is used when a model experiences frequent events. Example: A JSliderBar, passes many change events as the Grip is moved. Because the potential exists for so many events, the information in each is minimized.
- Stateful Notification - The event contains more state information than just the event source. Example: A JButton, passes just one action event as the button is pressed, so more information about the event is passed with each.

```
Model Interface                          Used by
                           Notification
BoundedRange Model                JProgressBar,JSlider,
JScrollBar                       Lightweight
Button Model                     JButton,   JCheckBox,
JCheckBoxMenuItem,          Stateful
JMenu, JMenuItem, Lightweight
JRadioButton, JRadioButtonMenuItem,
JToggleButton
ComboBox Model                    JComboBox
                    Stateful
```

10.3 JLabel

It used for display only, can display text, an image, or both. It cannot react to events, therefore cannot get the keyboard focus. Basically, it is an area where uneditable text and text with icons are displayed. You can set the